

TD1 - EDP.

Exo 1 : Problème en dim 1 sur $\Omega =]0,1[$

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{\partial^2 u}{\partial x^2}(t, x) = 0, & t > 0, x \in \Omega \\ u(t, x) = 0, & x \in \Gamma := \partial\Omega = \{0, 1\} \\ u(0, x) = u_0(x), & x \in \Omega \end{cases}$$

1.1) Soit $v \in C^1([0, 1])$, m.q.,
 $v(0) = 0$,

$$\int_0^1 v^2(x) dx \leq \int_0^1 \left(\frac{dv}{dx} \right)^2 dx$$

$$\text{Or : } v(x) = v(0) + \int_0^x v'(y) dy$$

(Rq : idem si $v \in H^1(]0, 1[)$, cf MIL et EDP 2)

$$\left(\int_0^x v'(y) dy \right)^2 \leq \left(\int_0^x v'(y) dy \right)^2 = \int_0^x (v'(y))^2 dy$$

$$C-S : |(x|y)| \leq \|x\| \cdot \|y\|$$

$$\text{Sur } L^2([0, 1]), (u|v)_{L^2} = \int_0^1 u(x) v(x) dx$$

$$\text{Q. a: } |v(x)| = \left| \int_0^x (v'(y)) dy \right|$$

$$\leq \int_0^x |v'(y)| dy$$

$$\leq \int_0^1 1 \cdot |v'(y)| dy$$

C-S:

$$|v(x)| \leq \left(\int_0^1 1^2 dy \right)^{1/2} \cdot \left(\int_0^x |v'(y)|^2 dy \right)^{1/2}$$

$$|v(x)|^2 \leq \int_0^x |v'(y)|^2 dy$$

$$\Rightarrow \int_0^1 |v(x)|^2 dx \leq \int_0^1 |v'(x)|^2 dx.$$

Req: sur $[0, L]$,

$$v(x) = v(0) + \int_0^x v'(y) dy$$

$$\downarrow$$

$$x \in [0, L]$$

$$\Rightarrow |v(x)| \leq \int_0^L |v'(y)| dy$$

$$\Rightarrow v \in L^\infty([0, L]) \text{ et}$$

$$\|v\|_\infty \leq \|v'\|_1$$

$$\leq C \|v'\|_2$$

$$\text{cf. } L^2([0, L]) \hookrightarrow L^4([0, L])$$

$$\text{car } \mu_2([0, L]) = L^2([0, L]) < \infty$$

(cf. M.J.1, D'Espaces L^p :

$$\|v\|_1 = \int_0^L 1 \cdot |v| dx = | (1 | v|) | \leq \underbrace{\|1\|_2}_{\sqrt{L}} \cdot \underbrace{\|v\|_2}_{\|v\|_2}$$

De m, $\exists \tilde{C} > 0$ tq:

$$\|v\|_2 \leq \tilde{C} \|v\|_\infty$$

(cf. $L^\infty([0, L]) \hookrightarrow L^2([0, L])$:

$$\int_0^L |v|^2 dx \leq \int_0^L \|v\|_\infty^2 dx = L \|v\|_\infty^2 < \infty$$

$$\Rightarrow v \in L^2([0, L]) \text{ et } \|v\|_2 \leq \sqrt{L} \cdot \|v\|_\infty$$

$$\begin{aligned} \text{Donc : } \|v\|_2 &\leq \tilde{C} \|v\|_\infty \\ &\leq \underbrace{\tilde{C}}_{\uparrow} \cdot C \|v\|_2 \\ &\quad (\sqrt{L})^2 = L \end{aligned}$$

1.2) Soit $u: \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$ sol de l'ÉDP de la chaleur, on définit l'énergie

$$E(t) := \int_0^1 u^2(t, x) dx$$

$$\bullet \text{ Eq: } \frac{1}{2} E'(t) = - \int_0^1 \left(\frac{\partial u}{\partial x}(t, x) \right)^2 dx$$

Rappel int. à paramètre (cf. M.J.1):

$$F(t) = \int_X f(t, x) d\mu(x)$$

i) $t \mapsto f(t, x)$ dérivable en t_0

ii) $f, g \in L^1(X, \mu, \mu)$ tq $|\frac{\partial f}{\partial t}(t, x)| \leq g(x)$, $x \in X$ et $t \in U \subset \mathbb{R}$

alors F bien définie & dérivable onto, et :

$$F'(t_0) = \frac{d}{dt} \int_X f(t_0, x) d\mu(x) \\ = \int_X \frac{\partial f}{\partial t}(t_0, x) d\mu(x).$$

. On suppose ici la régularité de la sol μ
suff pr app ce résultat :

E dérivable et :

$$E'(t) = \frac{d}{dt} \int_0^1 u^2(t, x) dx \\ = \int_0^1 \frac{\partial}{\partial t} (u^2(t, x)) dx \\ = \int_0^1 2 u(t, x) \cdot \frac{\partial u}{\partial t}(t, x) dx \\ = \int_0^1 2 u(t, x) \frac{\partial^2 u}{\partial x^2}(t, x) dx$$

$$= 2 \left(\int_0^1 u(t, x) \frac{\partial u}{\partial x}(t, x) dx - \int_0^1 u'(t, x) \frac{\partial u}{\partial x}(t, x) dx \right)$$

$$= 2 \left(0 - \int_0^1 u'(t, x) \frac{\partial u}{\partial x}(t, x) dx \right)$$

$$= -2 \int_0^1 \left(\frac{\partial u}{\partial x}(t, x) \right)^2 dx$$

$$\rightarrow 1 - \frac{1}{2} E'(t) = - \int_0^1 \left(\frac{\partial u}{\partial x} \right)^2 dx$$

$$1.3 \quad E(t) \leq E(0)e^{-2t}, \quad t \geq 0$$

$$\text{Or: } E'(t) = -2 \int_0^z \left(\frac{\partial u}{\partial x}(t, x) \right)^2 dx$$

$$\leq 0 \quad : \quad E' \leq 0$$

• Montrons de plus que:
 $E(t)e^{2t}$ est croissant $\rightarrow$ (auquel cas on a bien:

$$\rightarrow E(t) \cdot e^{2t} \geq E(0), \quad t \geq 0$$

$$\rightarrow E(t) \leq E(0) \cdot e^{-2t}$$

$$(E(t) \cdot e^{2t})' = \underbrace{E'(t)}_{\leq 0} e^{2t} + 2E(t)e^{2t}$$

$$= 2e^{2t} \underbrace{\left(E'(t) + 2E(t) \right)}_{\leq 0}$$

d'après 1.1)

$$\text{Or: } E(t) \rightarrow$$

(3)

$$\Rightarrow E(t)e^{2t} \leq E(0), \forall t \geq 0$$

$$\Rightarrow E(t) \leq E(0)e^{-2t}, \forall t \geq 0.$$

Exo 2: On considère l'éq. des ondes:

$$\left(\begin{array}{l} \frac{\partial^2 u}{\partial t^2}(t, x) - \frac{\partial^2 u}{\partial x^2}(t, x) = 0, t > 0, x \in \Omega \\ u(t, x) = 0, t > 0, x \in \Gamma := \partial\Omega \\ u(0, x) = u_0(x), x \in \Omega \\ \frac{\partial u}{\partial t}(0, x) = u_1(x), x \in \Omega \end{array} \right. \quad (\Omega \text{ ouvert } \subset \mathbb{R}^n)$$

On définit l'énergie selon

$$E(t) = \frac{1}{2} \int_{\Omega} \left(\left(\frac{\partial u}{\partial t}(t, x) \right)^2 + \left(\frac{\partial u}{\partial x}(t, x) \right)^2 \right) dx$$

• d'où l'énergie est préservée (avec u sol de l'éq.)

Or: Sous les bonnes hypothèses on a E dérivable

$$\textcircled{et} E'(t) = \frac{d}{dt} \left(\frac{1}{2} \int_{\Omega} \left(\left(\frac{\partial u}{\partial t} \right)^2 + \left(\frac{\partial u}{\partial x} \right)^2 \right) dx \right)$$

$$= \frac{1}{2} \int_{\Omega} \frac{d}{dt} \left[\left(\frac{\partial u}{\partial t} \right)^2 + \left(\frac{\partial u}{\partial x} \right)^2 \right] dx$$

$$= \int_{\Omega} \left[\frac{\partial u}{\partial t} \cdot \frac{\partial^2 u}{\partial t^2} + \frac{\partial u}{\partial x} \frac{\partial^2 u}{\partial t \partial x} \right] dx \quad \begin{array}{l} \nearrow = \frac{\partial^2 u}{\partial x^2} \text{ car u sol} \\ \searrow \end{array}$$

Cg.CL

2.2) En déduire l'unicité de sol. (NB: on ne dit rien de l'existence, voir p. cela 2.3 mais avec $\mathbb{R} = \mathbb{R}$).

→ Soit v_1 et v_2 2 sols de :

$$\left\{ \begin{array}{l} \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0, \quad t > 0, \quad u \in \mathcal{D}([0, 1] \times \mathbb{R}) \\ u(0, x) = u_0(x), \quad x \in \mathbb{R} \\ \frac{\partial u}{\partial t}(0, x) = u_1(x), \quad x \in \mathbb{R} \\ u(t, 0) = u(t, 1) = 0, \quad t > 0 \end{array} \right.$$

Par linéarité de l'EDP, $u := v_1 - v_2$ vérifie une (nouvelle) eq. des ondes:

$$\left\{ \begin{array}{l} \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0, \quad t > 0, x \in \mathbb{R} \\ \text{(cf. } \frac{\partial^2}{\partial t^2} (v_1 - v_2) = \frac{\partial^2 v_1}{\partial t^2} - \frac{\partial^2 v_2}{\partial t^2}, \text{ etc)} \\ u(0, x) = v_1(0, x) - v_2(0, x) = \cancel{u_0(x)} - u_d(x) \\ \qquad \qquad \qquad = u_0(x) - u_d(x) \\ \qquad \qquad \qquad = 0 \\ \frac{\partial u}{\partial t}(0, x) = u_1(x) - u_1(x) = 0 \\ u(t, 0) = v_1(t, \cancel{0}) - v_2(t, \cancel{0}) = 0 = u(t, 1), \quad t > 0 \end{array} \right.$$

$$\rightarrow \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial t} \right)$$

$\underbrace{\hspace{10em}}_{\frac{\partial^2 u}{\partial t \partial x}} \quad \underbrace{\hspace{10em}}_{\text{Schwarz} \quad \frac{\partial^2 u}{\partial x \partial t}}$

Qna:

$$\int_0^1 \frac{\partial u}{\partial x} \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial t} \right) dx$$

$$= \left[\frac{\partial u(t, x)}{\partial x} \frac{\partial u(t, x)}{\partial t} \right]_0^1 - \int_0^1 \frac{\partial^2 u}{\partial x^2} \frac{\partial u}{\partial t} dx$$

$$\left(\frac{\partial u(t, x)}{\partial x} \frac{\partial u(t, x)}{\partial t} - \frac{\partial u}{\partial t} \int_0^1 \frac{\partial^2 u}{\partial x^2} dx \right) (x)$$

$$\begin{cases} u(t, x) = 0, t > 0 \\ u(t, 0) = 0, t > 0 \\ u(t, 1) = 0 \end{cases}$$

$$\frac{\partial u}{\partial t}(t, 0) = \frac{\partial u}{\partial t}(t, 1) = 0$$

Dirichlet

$$\Rightarrow \epsilon'(t) = 0$$

$$\frac{\partial^2 u}{\partial t \partial x}$$

Qa: idem en faisant pp sur l'autre terme:

$$\int_0^1 \frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial x^2} dx = \left[\frac{\partial u}{\partial t} \frac{\partial u}{\partial x} \right]_0^1 - \int_0^1 \left| \frac{\partial u}{\partial x} \right| \frac{\partial u}{\partial t} dx$$

$$0 \leq t, E(0) = E(t)$$

$$\frac{1}{2} \int_0^1 \left[\left(\frac{\partial u}{\partial t}(0, x) \right)^2 + \left(\frac{\partial u}{\partial x}(0, x) \right)^2 \right] dx$$

$\circ$ $\frac{\partial u}{\partial t}(0, x)$ $\circ$ $\frac{\partial u}{\partial x}(0, x)$

$$\Rightarrow \frac{\partial u}{\partial t}(t, x) = \frac{\partial u}{\partial x}(t, x) = 0, t > 0, x \in \mathcal{L}$$

$$\rightarrow (\exists c \in \mathbb{R}) : u(t, x) = c = c_0 \text{ pour ex: } u(t, 0) = 0$$

$$\rightarrow v_1(t, x) = v_2(t, x), t > 0, x \in \mathcal{L}$$

2.3. $\mathcal{L} = \mathbb{R}$ (NB. $\mathcal{L}$ adaptés à ce cas :

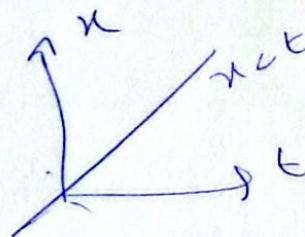
$$u(t, x) \rightarrow 0, \text{ cf. } \partial \mathcal{L} = \emptyset, |x| \rightarrow \infty$$

$$\text{Soit } u(t, x) = \frac{1}{2} \left(u_0(x+t) + u_0(x-t) \right) + \frac{1}{2} \left(v_1(x+t) - v_1(x-t) \right)$$

$$v_1'(x) = u_1(x)$$

$\swarrow$ onde rétrograde $\searrow$ onde progressive

Où u est sol de (*)



Où u est assez régulière $|x| = -k$
 dès que u_0 et u_1 le sont et :

$$\frac{\partial u}{\partial x}(t, x) = \frac{1}{2} \left[u_0'(x+t) + u_0'(x-t) \right] + \frac{1}{2} \left[\underbrace{v_1'(x+t)}_{u_1(x+t)} - \underbrace{v_1'(x-t)}_{u_1(x-t)} \right]$$

$$\frac{\partial^2 u}{\partial x^2}(t, x) = \frac{1}{2} \left[u_0''(x+t) + u_0''(x-t) \right] + \frac{1}{2} \left[\underbrace{v_1''(x+t)}_{u_1'(x+t)} - \underbrace{v_1''(x-t)}_{u_1'(x-t)} \right]$$

$$\left| \frac{\partial u}{\partial t}(t, x) \right| = \frac{1}{2} \left(u_0'(x+t) - u_0'(x-t) \right) + \frac{1}{2} \left(\underbrace{v_1'(x+t)}_{u_1(x+t)} + \underbrace{v_1'(x-t)}_{u_1(x-t)} \right)$$

$$\frac{\partial^2 u}{\partial t^2}(t, x) = \frac{1}{2} \left(u_0''(x+t) + u_0''(x-t) \right) + \frac{1}{2} \left(u_1'(x+t) - u_1'(x-t) \right)$$

→ Les CI st bien vérifiées et $\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$
 → $\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0$

Exo 3:

Exo 3:

$$u : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{C}$$

$$\begin{cases} i \frac{\partial u}{\partial t}(t, x) + \frac{\partial^2 u}{\partial x^2}(t, x) - V(x) u(t, x) = 0, \\ t > 0 \end{cases}$$

$$\begin{cases} u(0, x) = u_0(x), x \in \mathbb{R} \\ u(t, x) \xrightarrow{|x| \rightarrow \infty} 0, t > 0 \end{cases} \subset I$$

$$x \in \mathbb{R} = \mathbb{R}$$

$$\left. \begin{aligned} & \frac{\partial u}{\partial x}(t, x) \xrightarrow{|x| \rightarrow \infty} 0, t > 0 \\ & |u(t, x)|^2, \text{ densité} \end{aligned} \right\} \subset L \left(\begin{aligned} & n = \mathbb{R} \Rightarrow \\ & \text{cond de} \\ & \text{décroissance} \\ & \text{à } x \rightarrow \pm \infty \end{aligned} \right)$$

3.1) Soit $v : \mathbb{R}^n \rightarrow \mathbb{C} \simeq \mathbb{R}^2$, dérivable,

$$y \mapsto v(y) = v_1(y) + i v_2(y) = \begin{bmatrix} v_1(y) \\ v_2(y) \end{bmatrix}$$

$$\text{On a : } \frac{\partial}{\partial y_i} (|v(y)|^2)$$

$$= 2 \operatorname{Re} \left(\frac{\partial v}{\partial y_i}(y) \bar{v}(y) \right)$$

$$= 2 \operatorname{Re} \left(\frac{\partial \bar{v}}{\partial y_i}(y) \cdot v(y) \right)$$

$$\text{Or : } |v(y)|^2 = v_1^2(y) + v_2^2(y)$$

$$= v(y) \cdot \bar{v}(y)$$

$\mathbb{C}$, bilinéaire

$\Rightarrow |v|^2$ dérivable par comp (donc particulièrement dérivable)

et, si $i \in \{1, \dots, n\}$

$$\frac{\partial}{\partial y_i} (|v(y)|^2) = v'(y) \bar{v}(y) + v(y) (\bar{v}(y))'$$

$$= \frac{\partial v}{\partial y_i} \bar{v} + \overline{\frac{\partial v}{\partial y_i} v}$$

$$= 2 \operatorname{Re} \left(\bar{v} \frac{\partial v}{\partial y_i} \right)$$

$$= 2 \operatorname{Re} \left(v \cdot \frac{\partial \bar{v}}{\partial y_i} \right)$$

NB: En restant ds $\mathbb{R}^2$, on a de m :

$$|v(y)|^2 = v_1^2(y) + v_2^2(y) = \left\| \begin{pmatrix} v_1(y) \\ v_2(y) \end{pmatrix} \right\|^2$$

$$\left(\nabla (|v|^2) \right) = \frac{d}{dt} \left(v(y) |v'(y) \cdot h \right)$$

$$y \xrightarrow{v} v(y) \xrightarrow{\| \cdot \|} \mathbb{R}$$

$$\mathbb{R}^n \rightarrow \mathbb{R}^2 \rightarrow \mathbb{R} = \left(\underbrace{v'(y) \cdot v(t)}_{\nabla(|v|^2)} \right)$$

$$\Rightarrow \frac{\partial}{\partial y_i} (|v|^2)(y) = 2 \left(v_1(y) \frac{\partial v_1}{\partial y_i}(y) + v_2(y) \frac{\partial v_2}{\partial y_i}(y) \right)$$

et :

$$\operatorname{Re} \left(\frac{\partial v}{\partial y_i} \bar{v}(y) \right) = \frac{\partial v_1}{\partial y_i} v_1 - \frac{\partial v_2}{\partial y_i} (-v_2)$$

3.2) On définit, par μ_{sol} ,

$$E(t) = \frac{1}{2} \int_{\mathbb{R}} |\mu(t, x)|^2 dx$$

Où: l'énergie est conservée.

Or, ss les bonnes hypothèses, E est dérivable

$$\textcircled{et}: E'(t) = \frac{1}{2} \int_{\mathbb{R}} \frac{\partial}{\partial t} (|\mu(t, x)|^2) dx$$

$$= \frac{1}{2} \int_{\mathbb{R}} \text{Re} \left(\frac{\partial \mu}{\partial t} \bar{\mu} \right) dx$$

$$= \int_{\mathbb{R}} \text{Re} \left(\frac{\partial \mu}{\partial t} \bar{\mu} \right) dx$$

$$\rightarrow \frac{\partial \mu}{\partial t} = i \left(\frac{\partial^2 \mu}{\partial x^2} - V \mu \right) \quad \text{avec } -\frac{\partial^2 \mu}{\partial x^2} + V \mu = 0$$

$$\rightarrow \int_{\mathbb{R}} \text{Re} \left(i \bar{\mu} \frac{\partial^2 \mu}{\partial x^2} - i V \mu \bar{\mu} \right) dx$$

$$= \int_{\mathbb{R}} \text{Re} \left(i \bar{\mu} \frac{\partial^2 \mu}{\partial x^2} \right) dx$$

$$= \text{Re} \left(i \int_{\mathbb{R}} \bar{\mu} \frac{\partial^2 \mu}{\partial x^2} dx \right)$$

$$= \text{Re} \left(i \left(\left[\bar{\mu} \frac{\partial \mu}{\partial x} \right]_{\mathbb{R}} - \int_{\mathbb{R}} (\bar{\mu})' \frac{\partial \mu}{\partial x} dx \right) \right)$$

$$= \text{Re} \left(-i \int_{\mathbb{R}} (\bar{\mu})' \frac{\partial \mu}{\partial x} dx \right) = \text{Re} \left(-i \int_{\mathbb{R}} \frac{\partial}{\partial x} (\bar{\mu} \mu) dx \right)$$

→ on peut sortir Re pour intégrabilité

→

$$= - \int_{\mathbb{R}} \frac{\partial u}{\partial x} \times \frac{\overline{\partial u}}{\partial x} dx$$

$$= - \int_{\mathbb{R}} \left| \frac{\partial u}{\partial x} \right|^2 dx = 0$$

3.3) req:

$$F(t) := \int_{\mathbb{R}} \left[\left| \frac{\partial u}{\partial x}(t, x) \right|^2 + v(x) \cdot |u(t, x)|^2 \right] dx$$

et cste $\varphi \geq 0$. Sous les bonnes hypothèses, on peut comme précédemment dériver cette intégrale à paramètres (et) :

$$F'(t) = \int_{\mathbb{R}} \frac{\partial}{\partial t} \left[\left| \frac{\partial u}{\partial x} \right|^2 + v(x) |u|^2 \right] dx$$

$$\begin{aligned} & \stackrel{3.1}{=} 2 \int_{\mathbb{R}} \operatorname{Re} \left(\underbrace{\frac{\partial^2 u}{\partial t \partial x} \cdot \frac{\overline{\partial u}}{\partial x}}_{\frac{\partial^2 \bar{u}}{\partial t \partial x} \cdot \frac{\partial u}{\partial x}} + v(x) \underbrace{\frac{\partial u}{\partial t} \bar{u}}_{\frac{\partial \bar{u}}{\partial t} \cdot u} \right) dx \\ & = \frac{\partial^2 \bar{u}}{\partial t \partial x} \cdot \frac{\partial u}{\partial x} \end{aligned}$$

$$\text{Or, } u \text{ sol} \Rightarrow i \frac{\partial u}{\partial t} + \frac{\partial^2 u}{\partial x^2} - v u = 0$$

$$\times \frac{\overline{\partial u}}{\partial t} : \left| \frac{\partial u}{\partial x} \right|^2 + \left(\frac{\partial^2 u}{\partial x^2} - v u \right) \frac{\partial u}{\partial t} = 0$$

$\dot{\Sigma} t$ on a , par i p p,

$$\int_{\mathbb{R}} \frac{\partial^2 \bar{u}}{\partial x \partial t} \frac{\partial u}{\partial x} dx$$

$$= \left[\frac{\partial u}{\partial x} \frac{\partial \bar{u}}{\partial t} \right]_{-\infty}^{+\infty}$$

$$- \int_{-\infty}^{+\infty} \frac{\partial^2 u}{\partial x^2} \frac{\partial \bar{u}}{\partial t} dx$$

$$= - \int_{-\infty}^{+\infty} \frac{\partial \bar{u}}{\partial t} \frac{\partial^2 u}{\partial x^2} dx$$

$$F'(t) = 2 \operatorname{Re} \left(\int_{\mathbb{R}} \left[-\frac{\partial \bar{u}}{\partial t} \frac{\partial^2 u}{\partial x^2} + v \frac{\partial \bar{u}}{\partial t} u \right] dx \right)$$

$$\left(-\frac{\partial^2 u}{\partial x^2} + V u \right) \frac{\partial \bar{u}}{\partial t} = i \left| \frac{\partial u}{\partial t} \right|^2$$

$$= 0$$